Inference Problems

Monday, October 19, 2026 2:00 PM

Problem 1:

Sample mean $\bar{X} = 52$

Sample standard deviation $s = 10$

$n = 100$

Hypothesized population mean $\mu_0 = 50$

- Calculate the estimated standard error

$$SE(\bar{X}) = \frac{s}{\sqrt{n}}$$

$$= \frac{10}{\sqrt{100}} = \frac{10}{10} = \boxed{1}$$

- Hypothesis Test

$H_0 : \mu = 50$ ← Two sided test

$H_1 : \mu \neq 50$

$$z = \frac{\bar{X} - \mu_0}{s / \sqrt{n}} = \frac{52 - 50}{10 / \sqrt{100}} = \frac{2}{1} = 2$$

	Significance level	Critical Value	Comparison	Decision
90%	$\alpha = 0.10$	1.645	$2 > 1.645$	Reject H_0
95%	$\alpha = 0.05$	1.96	$2 > 1.96$	Reject H_0
99%	$\alpha = 0.01$	2.576	$2 < 2.576$	Fail to reject H_0

- We reject H_0 at significance levels 0.10 & 0.05, but fail to reject at 0.01. This means that we have sufficient statistical evidence that the population mean differs from 50 at 10% & 5% significance levels, but not at 1%.

- 95% confidence interval for μ

large-sample Normal confidence interval:

$$\bar{X} \pm z^* \left(\frac{s}{\sqrt{n}}\right) \qquad SE = \frac{s}{\sqrt{n}} = \frac{10}{\sqrt{100}} = 1$$

↑ center of interval

↑ critical value associated with confidence interval

estimated standard error SE

$52 \pm 1.96(1)$

$52 \pm 1.96 \rightarrow \boxed{(50.04, 53.96)}$

↳ 95% confident that the true pop. mean is in this interval

Problem 2:

Sample estimate $\hat{S} = 3.1$

Estimated standard error $\widehat{SE}(\hat{S}) = 1.3$

Value testing against $S_0 = 0$

- Confidence Intervals

Estimate $\pm z^* \times SE$

$\hat{S} \pm z^* \widehat{SE}(\hat{S})$

$3.1 \pm z^* (1.3$

90% CI: $z^* = 1.645$

$3.1 + 1.645(1.3) = 5.2385$

$3.1 - 1.645(1.3) = 0.9615$

$\boxed{90\%\ CI = (0.9615, 5.2385)}$

95% CI: $z^* = 1.96$

$3.1 + 1.96(1.3) = 5.648$

$3.1 - 1.96(1.3) = 0.552$

$\boxed{95\%\ CI = (0.552, 5.648)}$

99% CI: $z^* = 2.576$

$3.1 + 2.576(1.3) = 6.4488$

$3.1 - 2.576(1.3) = -0.2488$

$\boxed{99\%\ CI = (-0.2488, 6.4488)}$

- Hypothesis Test

$H_0 : S = 0$

$H_1 : S = 1$

$$z = \frac{\text{Estimate} - \text{Null Value}}{\text{Standard error}} = \frac{3.1 - 0}{1.3} = 2.385$$

	Significance level	Critical Value	Comparison	Decision
90%	$\alpha = 0.10$	1.645	$2.385 > 1.645$	Reject H_0
95%	$\alpha = 0.05$	1.96	$2.385 > 1.645$	Reject H_0
99%	$\alpha = 0.01$	2.576	$2.385 < 2.576$	Fail to reject H_0

- At $\alpha = 0.10$, reject H_0
- At $\alpha = 0.05$, reject H_0
- At $\alpha = 0.01$, fail to reject H_0

Problem 3:

- Sample mean $\bar{X} = 11.5$

$s = 3$

$n = 36$

$\mu_0 = 12$

$z^* = 1.96$

- $SE(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{3}{\sqrt{36}} = \frac{3}{6} = \frac{1}{2}$
- 95% CI:

$\bar{X} \pm z^* \cdot SE$

$11.5 + 1.96(0.5) = 12.48$

$11.5 - 1.96(0.5) = 10.52$

$\boxed{95\%\ CI = (10.52, 12.48)}$

↳ Yes, 12 is a plausible value for the population mean because it lies in this interval.

- Hypothesis Test

$H_0 : \mu = 12$

$H_1 : \mu \neq 12$

$$z = \frac{\bar{x} - \mu_0}{SE} = \frac{11.5 - 12}{0.5} = -1$$

$\boxed{\text{Since } |-1| = 1 < 1.96, \text{ we fail to reject } H_0 \text{ at } \alpha = 0.05}$

Problem 4

	Group A	Group B
$\bar{X}$	25	22
s	8	7
n	100	100

$H_0 : \mu_A - \mu_B = 0$

$H_1 : \mu_A - \mu_B \neq 0$

$$SE(\bar{x}_A - \bar{x}_B) = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = \sqrt{\frac{8^2}{100} + \frac{7^2}{100}} = \sqrt{\frac{64}{100} + \frac{49}{100}}$$

$$= \sqrt{0.64 + 0.49} = 1.063$$ ← SE of difference between the two samples

Z statistic:

$$z = \frac{(\bar{x}_A - \bar{x}_B) - 0}{SE} = \frac{(25 - 22) - 0}{1.063} = \boxed{2.822}$$

- Since $2.822 > 1.96$, we reject H_0 at $\alpha = 0.05$

95% CI:

$(\bar{x}_A - \bar{x}_B) \pm z^* \cdot SE$

$3 + 1.96(1.063) = 5.084$

$3 - 1.96(1.063) = 0.916$

$\boxed{95\%\ CI = (0.916, 5.084)}$

Problem 5:

$\bar{X}_A = 14$, $s_A = 6$, $n_A = 81$

$\bar{X}_B = 12$, $s_B = 5$, $n_B = 64$

- SE:

$$SE(\bar{x}_A - \bar{x}_B) = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = \sqrt{\frac{6^2}{81} + \frac{5^2}{64}} = \sqrt{\frac{36}{81} + \frac{25}{64}}$$

$\boxed{SE = 0.914}$

- z statistic:

$$z = \frac{(\bar{x}_A - \bar{x}_B) - (\mu_A - \mu_B)_0}{SE}$$

↳ $H_0 : \mu_A - \mu_B = 0$

$$z = \frac{(14 - 12) - 0}{0.914} = \frac{2}{0.914} = \boxed{2.189}$$

- At $\alpha = 0.05$, we reject H_0 bc $2.189 > 1.96$

Problem 6

$\bar{X}_A = 3.2$, $s_A = 1.5$, $n_A = 49$

$\bar{X}_B = 2.8$, $s_B = 1.4$, $n_B = 64$

$H_0 : \mu_A - \mu_B = 0$

- SE:

$$SE = \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = \sqrt{\frac{1.5^2}{49} + \frac{1.4^2}{64}} = 0.2767$$

- z statistic

$$z = \frac{(\bar{x}_A - \bar{x}_B) - 0}{SE} = \frac{(3.2 - 2.8) - 0}{0.2767}$$

$\boxed{z = 1.446}$

- Test at $\alpha = 0.05$:

$z^* = 1.96$

$|z| > 1.96 \rightarrow$ Reject H_0

$1.446 < 1.96$

↳ fail to reject H_0

- 95% CI :

$(\bar{x}_A - \bar{x}_B) \pm z^* \cdot SE$

$0.4 + 1.96(0.2767) = -0.1423$

$0.4 - 1.96(0.2767) = 0.9423$

$\boxed{95\%\ CI = (-0.1423, 0.9423)}$

- The 0.4 point gap in defect scores is not statistically significant at this sample size. It is small relative to the noise.